

**Managing Wildfire Risk After Mountain Pine Beetle:
Evaluating Strategies in British Columbia's Changing Forests**

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1.0 Introduction

Mountain pine beetle (MPB) and wildfire are two of the most influential natural disturbances in western North America's lodgepole pine forests. These processes have long played a role in shaping forest structure and regeneration. However, over the past several decades, both have intensified due to widespread fire suppression and increasing temperatures. Warmer winters, longer drought and dense forest stands have allowed MPB populations to reach outbreak levels across large areas (Jenkins et al., 2014). This is particularly true in British Columbia (BC), where more than 18 million hectares have been affected since the 1990s (Dhar et al., 2016). At the same time, wildfire activity has increased in both frequency and severity (Woo et al., 2024), creating concern about the long-term forest resilience.

MPB outbreaks significantly alter forest structure and fuel conditions, affecting wildfire behavior and complicating management (Jenkins et al., 2014). As beetle killed trees go through different post-infestation stages, fuels shift in type, amount and arrangement, leading to variable fire risks over time (Schoennagel et al., 2011; Talucci et al., 2022). These shifting fuel conditions pose challenges for wildfire management, particularly as fire seasons become longer and more intense across BC.

Despite the overlap between beetle outbreaks and wildfire risk, these disturbances are often treated separately in management. Although these disturbances are linked, fire and beetle management have typically been approached as separate issues with different objectives, timelines and tools guiding decisions on each side. Research in BC shows that these disturbances are closely linked, and that future management could benefit from more integrated planning (Woo et al., 2024).

This paper explores how wildfire risk evolves over time following mountain pine beetle outbreaks and evaluates the use of three forest management strategies in reducing that risk: prescribed burning, mechanical thinning and salvage logging. Using examples from British Columbia, it draws on two case studies: the Mitchell Ridge prescribed burn and the Central Chilcotin Rehabilitation Ltd.'s Indigenous-led forest recovery initiatives. These examples help illustrate how fire risk changes under different post-beetle conditions and how factors like local context and timing shape the effectiveness of management strategies.

In doing so, this paper not only evaluates specific management strategies but also engages with a broader more complex question: How do we achieve balance between actively managing these ecological processes and knowing when to let them play out on their own?

2.0 Literature Review

2.1 MPB and Fire Risk Over Time

MPB outbreaks have significant long-term impacts on forest structure and fuel dynamics, influencing fire behavior in complex ways. As beetle killed trees transition through stages of decay, the type and arrangement of fuels change, altering wildfire risk over time. These changes affect fire behaviour differently depending on factors such as time since outbreak, weather conditions, and local forest characteristics (Jenkins et al., 2014; Schoennagel et al., 2012). MPB affected forests are usually divided into four stages: green, red, grey, and old-MPB. These stages and their impacts on fuels and fire behaviour are summarized visually in Figure 1 (from Hicke et al., 2012).

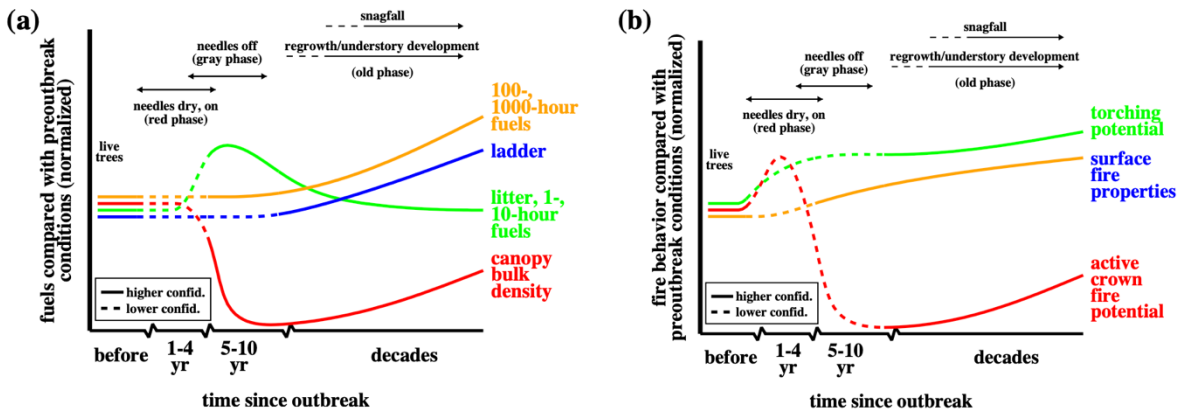


Figure 1. Model showing changes in (a) fuel characteristics and (b) fire behaviour following MPB outbreaks across different stages: Green (pre-outbreak), Red(1-4yr), Grey(5-10yr), and Old(10+yr). The figure illustrates trends in canopy bulk density, surface fuels, ladder fuels and fire potential over time. Figure from Hicke et al. (2012).

In the green stage, beetles are not yet active, but stand density is often high and many trees are already vulnerable due to competition and stress. While this stage presents no MPB specific fuel changes, the stands are still prone to high-intensity wildfire under dry conditions (Jenkins et al., 2014; Hicke et al., 2012).

The red stage, occurring one to three years post-infestation, is characterized by recently killed trees that still retain dry, red needles. Foliar moisture drops significantly during this period, increasing crown fire potential. Studies have shown that even under moderate weather, red-stage stands can support active crown fire, particularly under wind-driven conditions (Jenkins et al., 2014; Schoennagel et al., 2012; Hicke et al., 2012).

In the grey stage (four to ten years after infestation), needles have fallen, fine branches begin to break off and snags start collapsing. Canopy fuels are reduced, but heavy surface fuels accumulate and ladder fuels emerge from regeneration and deadfall. These conditions elevate the

risk of intense surface fires, especially during hot, dry weather (Schoennagel et al., 2012; Talucci et al., 2022; Hicke et al., 2012). Additionally, more open canopies allow greater wind and sunlight penetration, promoting faster fuel drying (Schoennagel et al., 2012).

By the old-MPB stage (ten or more years after infestation), most snags have fallen, and a dense understory of regenerating trees is often established. Surface fuels remain high, and the regenerating understory creates conditions favourable for fire to move back into the canopy or ignite remaining standing dead trees. Talucci et al. (2022) found that fire severity often peaks during this stage, especially where a mix of live and dead vegetation increases flammability. This pattern is consistent with the fuel structure described by Hicke et al. (2012).

While some U.S. based studies argue that fire weather is the dominant driver of fire behaviour regardless of MPB presence (Meigs et al., 2016), research in British Columbia paints a more complex picture. In BC, MPB affected areas have been shown to burn more often under lightning ignition and severe weather conditions, pointing to a stronger link between beetle disturbance and large fires (Woo et al., 2024). This likely reflects BC's extensive lodgepole pine forests, which are highly susceptible to beetle outbreaks (Dhar et al., 2016), coupled with heavy beetle induced fuel loads, increasing drought conditions and decades of fire suppression that have allowed fuels to accumulate (Jenkins et al., 2014; Woo et al., 2024).

These findings highlight that wildfire risk following MPB is dynamic and location dependent. Shifts in fuel structure, time since outbreak and forest conditions all shape fire behaviour differently. This variability makes standardized approaches less effective. Instead, adaptive, site-specific strategies are needed to manage long-term fire risk in beetle-affected forests.

2.2 Management Strategies for Reducing Fire Risk

Fire risk in MPB-affected forests shifts over time, and management strategies must adapt to these changing conditions. Three common approaches—prescribed burning, mechanical thinning, and salvage logging—are used to reduce fuel loads and mitigate wildfire hazards. Each strategy has strengths and limitations. Their effectiveness depends heavily on when, where and how they are applied.

Prescribed Burning

Prescribed fire is most commonly used in the grey stage of MPB disturbance, when needles have fallen and surface fuels begin to build up. At this point crown fire risk may be lower, but deadfall and regenerating understory create a high potential for intense surface and ladder fires (Talucci et al., 2022). Prescribed burning is often used to reduce these fuels and help mimic natural fire regimes in fire-adapted landscapes. However, fire behaviour in beetle affected stands can be difficult to predict, especially when dry snags or downed material are present. In some areas, prescribed burning may also be limited by operational constraints or public concerns for safety. Some studies have noted that fire may increase tree stress and beetle vulnerability, particularly in areas where beetle populations are still active and nearby trees remain susceptible to infestation (Jenkins et al., 2014).

Mechanical Thinning

Mechanical thinning is often used in the early stages of beetle outbreaks, especially in dense stands that are highly susceptible to infestation. Thinning reduces competition among trees, which improves vigor and resistance to beetle attack (Jenkins et al., 2014). It also lowers crown bulk density and removes ladder fuels, helping to reduce the potential for crown fire. When paired with prescribed fire, thinning is especially effective in reducing overall fire hazard (Fettig

et al., 2021). However, its effectiveness depends on how and when it is implemented. If thinning leaves too much untreated slash, it can increase surface fuel loads and create new ignition risks. In some cases, logging residue may even attract beetles if not managed properly (Fettig et al., 2021). Thinning is rarely used once a stand is already heavily affected by MPB, but it can still be useful in partially infested stands or as a preparatory step for other treatments.

Salvage Logging

Salvage logging is often used after MPB outbreaks to remove dead and dying trees, recover timber value, and reduce fuel loads. It is one of the most visible management responses in BC following large-scale beetle mortality. From a fire risk perspective, salvage can help reduce standing dead material and create access for other treatments. However, its effectiveness depends on how the operation is carried out. If slash is left on site or not properly treated, surface fuels can actually increase, raising the potential for high-intensity surface fires (Dhar et al., 2016). Ecologically, salvage logging is controversial. Large-scale operations can compact soils, disrupt natural regeneration and reduce structural diversity, potentially affecting long-term recovery. The removal of standing dead trees can also eliminate important wildlife habitat. As a result, some researchers argue that more selective or small-scale salvage approaches may offer a better balance between reducing fire hazard and maintaining forest resilience (Dhar et al., 2016).

Integration

These strategies are often applied independently, although combining treatments can improve outcomes. For example, thinning followed by prescribed burning can target both ladder and surface fuels, reducing overall flammability more effectively than either method alone (Jenkins et al., 2014; Woo et al., 2024). MPB affected forests often contain a complex mix of dead trees, fallen material and regenerating vegetation, which creates challenging fuel conditions over time

(Talucci et al., 2022). Integrated and adaptive approaches that are tailored to stand condition and timing, are likely to be more successful in reducing fire risk than single strategies applied in isolation.

2.3 Case Studies: Prescribed Burning and Salvage Logging in British Columbia

Real-world examples offer insight into how MPB related fire risk is managed under different ecological, cultural, and operational conditions. In British Columbia, two case studies — the Mitchell Ridge prescribed burn and Central Chilcotin Rehabilitation Ltd. can help illustrate how timing, objectives and local context influence the effectiveness of management strategies.

Mitchell Ridge Prescribed Burn

The Mitchell Ridge prescribed burn took place in Kootenay National Park in southeastern BC as part of Parks Canada's effort to reintroduce fire into fire-adapted landscapes affected by mountain pine beetle. The area had experienced significant MPB related lodgepole pine mortality. The burn was designed to meet several objectives: reduce beetle-susceptible habitat, lower the risk of high-intensity wildfire and support ecological restoration (Kubian, 2010).

In 2009, three adjacent plots were burned under similar weather and fuel moisture conditions: one with minimal beetle impact, one in the red stage, and one in the grey stage. Fire intensity and surface fuel consumption were highest in the grey-stage plot, where deadfall and fine fuels had accumulated. However, crown fire occurred in all three plots, including the one with minimal beetle activity. This showed that MPB presence alone does not determine crown fire potential (Kubian, 2010).

This case supports the idea that the grey stage may be a practical window for prescribed burning, when surface fuels are elevated but widespread snag fall and dense regeneration have

not yet made conditions more hazardous. It also demonstrates how fire behaviour changes as MPB disturbance progresses, and why treatment timing is critical in reducing fire risk. When implemented under the right conditions, prescribed fire can be an effective tool for managing fuels in beetle-affected forests and supporting broader fire resilience strategies.

Central Chilcotin Rehabilitation Ltd.: Indigenous-Led Forest Recovery

Central Chilcotin Rehabilitation Ltd. (CCR) is a First Nations-owned company formed in 2017 as a joint venture between the T̓sideldel First Nation and the Tl'etinqox Government. Established in response to widespread MPB mortality and the 2017 wildfire season. CCR integrates salvage logging, wildfire risk reduction and reforestation while centering Indigenous values and community leadership (Canadian Geographic, 2024; Central Chilcotin Rehabilitation Ltd., n.d.).

CCR's projects focus on removing beetle killed and fire damaged timber to reduce fuel loads and make use of dead wood that would otherwise go to waste. This helps support local mills and bioenergy facilities while lowering fire risk across the landscape. By focusing on strategic salvage and ecosystem restoration, CCR aims to mitigate future fire severity while facilitating long-term forest recovery. Replanting efforts have included drone assisted seeding in severely burned stands where natural regeneration is limited, helping accelerate regeneration in difficult terrain (Canadian Biomass, 2024).

This case highlights how post-MPB salvage can be adapted to meet both ecological and community goals. CCR demonstrates that when salvage logging is integrated with reforestation and grounded in Indigenous knowledge, it can contribute to long-term forest resilience and wildfire risk reduction in beetle-affected landscapes.

2.4 Coordinating Management vs. Letting Nature Work

The effects of mountain pine beetle outbreaks and wildfire are complex, long-term and often unpredictable. As this paper has shown, fire risk in MPB-affected forests is dependent on timing and site conditions. Management tools like prescribed burning, thinning, and salvage logging can reduce this risk, but their outcomes vary depending on when and where they are used. These strategies all come with trade-offs—ecologically, socially, and financially—and relying on them in isolation may limit their impact.

The Mitchell Ridge and CCR examples show how different approaches to intervention can reflect local conditions and goals. The Mitchell Ridge prescribed burn reduced fuels under controlled conditions, while CCR’s salvage operations removed dead timber and supported replanting efforts. Both examples demonstrate that proactive strategies can be effective, but only when well-timed and appropriate to the forest’s condition.

This raises a broader question: how do we decide when to intervene, and when might it be more appropriate to step back? In some forests, particularly where conditions are less prone to severe fire or recovery is already underway, natural processes may be sufficient. In others, especially where fuel accumulation is high and fire weather conditions are intensifying, inaction could lead to more severe outcomes. Recognizing when a forest can recover on its own and when that may lead to greater risk is essential to designing effective, long-term strategies.

Finding balance means acknowledging that no single strategy will work in all situations. Coordinated planning that considers both beetle and fire dynamics and that remains flexible as conditions change, will be key. In some forests, active management may be needed. In others, stepping back and allowing natural processes to unfold could be the better path.

3.0 Summary & Implications

Mountain pine beetle outbreaks have significantly altered fire risk dynamics across many forests in British Columbia. As trees transition through different post-infestation stages, fuel structures shift in ways that can increase the likelihood and severity of wildfire. This fire risk is not static. It evolves over time and varies based on local conditions. Understanding these patterns is essential for developing management strategies that are both effective and adaptive.

This paper evaluated three common approaches to managing fire risk in MPB affected forests: prescribed burning, mechanical thinning, and salvage logging. Each strategy can reduce fuels and support forest resilience, but their effectiveness depends heavily on when, where and how they are applied. Timing, site conditions and the stage of beetle disturbance all influence outcomes. Prescribed burning appears most effective during the grey stage, when surface fuels are elevated but regeneration is not yet dense. Thinning is best suited to early intervention, particularly in vulnerable stands that have not yet experienced full mortality. Salvage logging offers economic and operational benefits, but its ecological impacts and fuel-related outcomes are highly variable. The Mitchell Ridge and CCR case studies showed how these management strategies can be tailored to local goals and conditions.

The literature and case studies make it clear that no single strategy works everywhere. In some forests, active management may be necessary to reduce fuel loads and support recovery. In others, allowing natural processes to unfold, such as low-severity wildfires or gradual regeneration, may be a more appropriate approach. The right approach depends on context.

MPB and wildfire, however, are still often managed as separate issues despite their close connection. This reflects a disconnect in research and practice: while beetle outbreaks and wildfire risk are clearly connected, they have often been addressed in isolation. More integrated approaches are needed. Future forest and fire management in BC should consider both beetle

dynamics and fire behaviour when designing treatments. This could include combining treatments such as thinning and prescribed fire, coordinating fire and beetle management plans, and adapting strategies to local fuel conditions, forest structure and community priorities.

Looking ahead, long-term forest resilience in British Columbia will depend on our ability to manage disturbance in a way that is flexible, locally informed and forward thinking. That includes recognizing that mountain pine beetle is a natural part of these ecosystems. The goal is not to eliminate beetles entirely, but to reduce the severity of future outbreaks in ways that also lower wildfire risk. Strategies like prescribed fire or selective logging may not directly kill beetles, but they can disrupt their habitat and life cycle while reducing fuels. In some cases, letting natural disturbances unfold may be appropriate. In others, carefully planned management can support forest recovery and minimize compounding impacts over time. The challenge will be knowing when to step in and when to let natural processes guide the recovery.

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